Test 1

Numerical Methods, Computer Science BSc

April 13, 2023

Solve the below problems! Write your solution of each problem on a separate page.

1. Consider the set of machine numbers $M = M(5, -6, 6)$.

(a) Identify $\varepsilon_0, \varepsilon_1$ and M_∞ in this set.

(b) Calculate the machine numbers that correspond to 0.27 and 3.8.

(c) Perform the machine addition $fl(0.27) \oplus fl(3.8)$.

(d) Provide an absolute error bound for $fl(0.27)$, $fl(3.8)$ and their sum.

(3+3+3+3 points)

2. Calculate the absolute error bound for the value

$$\frac{\sqrt{a \cdot b + b \cdot b}}{b}$$

if $a = 6$ and $b = 3$ are approximations with $\Delta_a = \frac{5}{3}$ and $\Delta_b = \frac{1}{2}$. *(8 points)*

3. Let

$$A = \begin{bmatrix} 2 & -4 & 1 \\ -2 & -1 & 3 \\ 4 & -2 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} -9 \\ 3 \\ -4 \end{bmatrix}.$$

(a) Solve the system of linear equations $Ax = b$ using Gaussian elimination with **partial pivoting**.

(b) Find $\det(A)$ from the previous results.

(c) Give the LU decomposition of A. (Should not use the above results!)

(5+1+4 points)

4. Calculate $\mathrm{cond}_\infty(A)$ and $\mathrm{cond}_2(A)$ with respect to the matrix

$$A = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 4 & -2 & 2 \end{bmatrix}.$$

(5+5 points)

5. Consider the polynomial

$$P(x) = x^4 - 4x^3 - x^2 + 16x - 12.$$

(a) Give lower and upper bounds for the absolute values of the roots.

(b) Compute the Taylor polynomial of P centered at $a = 2$ using Horner's method.

(4+6 points)

Good work!